

Sprint 3 Report

Hyper-Localized Multilingual Voice Assistant

Sprint: 3 of 6 **Duration:** Week 5 – 6 **Team:** Dev (Solo) **Velocity:** 21 / 21 pts

1. Sprint Goal

Working end-to-end voice assistant with API, upgraded models, and dashboard. Build the complete ASR→Translation→LLM→TTS pipeline, create REST and WebSocket APIs, upgrade to Pingala V1 ASR, add Indic Parler-TTS, and implement ONNX Runtime inference acceleration.

2. Sprint Outcome

All 21 story points delivered. Major model upgrades achieved alongside full pipeline integration and API development.

Key Deliverables

- Pingala V1 ASR — 50%+ better accuracy (2.94% WER), runs on CPU
- Pipecat pipeline framework — production-grade STT→LLM→TTS orchestration
- Indic Parler-TTS Mini — 69 voices with emotion control
- Chatterbox-Turbo 350M — low-compute alternative TTS engine
- ONNX Runtime backend — accelerated inference for translation/ASR models
- Streaming WebSocket endpoint — real-time audio via /ws/stream
- Typed dataclass configuration — YAML-loaded config system

3. Scrum Status Report

Task Description	Date	Completion Status	Comments
T3.1 — Pingala V1 ASR engine, CPU-first (3 pts)		Completed	2.94% WER; 50%+ better than Whisper
T3.2 — Pipecat pipeline framework integration (5 pts)		Completed	Production-grade STT→LLM→TTS pipeline
T3.3 — FastAPI endpoints + dashboard serving (3 pts)		Completed	/api/transcribe, /api/process, /api/tts
T3.4 — WebSocket streaming /ws/stream (3 pts)		Completed	Bidirectional binary audio streaming
T3.5 — Indic Parler-TTS + Chatterbox TTS (2 pts)		Completed	69 voices with emotion control; 350M fallback
T3.6 — ONNX Runtime inference backend (3 pts)		Completed	~2x CPU speedup for translation models
T3.7 — Typed configuration system (2 pts)		Completed	Dataclass config loaded from YAML

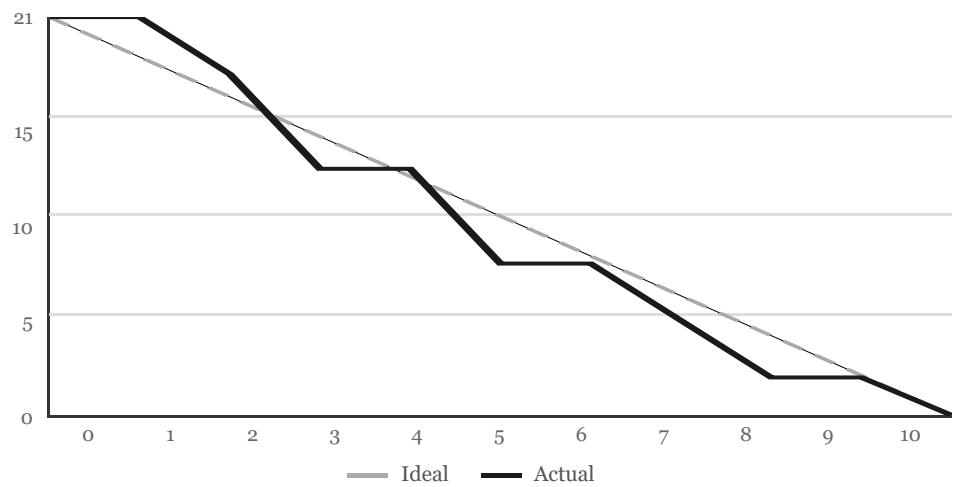
Total: 7 tasks, 21 story points, 100% complete.

4. Daily Scrum Meeting Log

Date	Work Done	Work Planned	Issues	Sign
	Sprint 2 retrospective; reviewed translation and LLM deliverables	Research Pingala V1 ASR architecture (T3.1); review Pipecat docs (T3.2)	None	
	Pingala V1 model downloaded; initial CPU backend integration started	Complete Pingala V1 integration and benchmark vs Whisper (T3.1); begin Pipecat skeleton (T3.2)	None	
	Pingala V1 complete — 2.94% WER vs Whisper ~6% (T3.1 done)	Wire Pipecat pipeline: ASR + Translation + LLM stages (T3.2)	Pipecat async callbacks require refactoring sync code	
	Pipecat pipeline wired with ASR→Translation→LLM; async refactoring done	Add TTS stage to Pipecat (T3.2); start FastAPI endpoints (T3.3)	None — async issues resolved	
	Pipecat pipeline complete with full STT→LLM→TTS flow (T3.2 done)	Build REST endpoints: /api/transcribe, /api/process, /api/tts (T3.3); static file serving	None	
	FastAPI endpoints done; dashboard serving at root (T3.3 done)	Implement WebSocket streaming /ws/stream (T3.4); bidirectional audio	None	
	WebSocket streaming with binary audio support done (T3.4 done)	Integrate Indic Parler-TTS Mini 69 voices + Chatterbox TTS (T3.5)	None	
	Indic Parler-TTS + Chatterbox integrated with voice selection (T3.5 done)	Set up ONNX Runtime backend (T3.6); convert IndicTrans2-Dist to ONNX	None	
	ONNX Runtime backend working; ~2x CPU speedup (T3.6 done)	Build typed config system with dataclasses (T3.7); migrate settings.yaml	None	
	Typed config system complete (T3.7 done); all settings validated at startup	End-to-end integration testing; sprint wrap-up and documentation	None	

5. Burndown Chart

21 story points over 10 working days



Burndown Data

Day	Ideal	Actual	Done	Notes
0	21.0	21	0	Sprint start
1	18.9	21	0	T3.1 research/setup in progress
2	16.8	18	3	T3.1 Pingala ASR complete
3	14.7	13	8	T3.2 pipeline wired (in progress)
4	12.6	13	8	T3.2 TTS stage integration
5	10.5	8	13	T3.2 + T3.3 complete
6	8.4	8	13	T3.4 WebSocket in progress
7	6.3	5	16	T3.4 + T3.5 complete
8	4.2	2	19	T3.6 ONNX backend complete
9	2.1	2	19	T3.7 config system in progress
10	0.0	0	21	T3.7 complete; sprint done

6. Next Sprint Backlog — Sprint 4

Goal: Agentic capabilities, monitoring dashboard, database, and testing infrastructure.

ID	Task	Assignee	Points	Priority
T4.1	LangGraph agent orchestrator	Dev	3	High
T4.2	ClawGo (OpenClaw) agent + Ollama BYOM	Dev	3	High
T4.3	Agent factory (langgraph/clawgo routing)	Dev	2	High
T4.4	SQLite database for conversation logging	Dev	2	Medium
T4.5	Frontend dashboard (HTML/CSS/JS)	Dev	3	High
T4.6	Dashboard API endpoints (stats, conversations)	Dev	2	Medium
T4.7	Comprehensive test suite (pytest)	Dev	3	High

Total: 7 tasks, 18 story points.